

6

International Trade and Investment



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LEARNING OBJECTIVES

- 6.1. Understand the motivation for international trade.
- 6.2. Summarize and discuss the differences among the classical country-based theories of international trade.
- 6.3. Use the modern firm-based theories of international trade to describe global strategies adopted by businesses.
- 6.4. Describe and categorize the different forms of international investment.
- 6.5. Explain the reasons for foreign direct investment.
- 6.6. Summarize how supply, demand, and political factors influence foreign direct investment.

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FloraHolland and its members are prime examples of the firms throughout the world who have used technology, knowledge, and the resources at their disposal to compete successfully in the global marketplace. In this chapter we analyze the underlying economic forces that shape and structure the international business transactions conducted by firms in the floriculture industry and in thousands of other industries. We discuss the major theories that explain and predict international trade and investment activity. These theories help firms sharpen their global business strategies, identify promising export and investment opportunities, and react to threats posed by foreign competitors. As they introduce you to the economic environment in which firms compete, the theories help you understand why Dutch floriculture firms, despite their high labor costs, have proved to be so successful in international markets.

International Trade and the World Economy

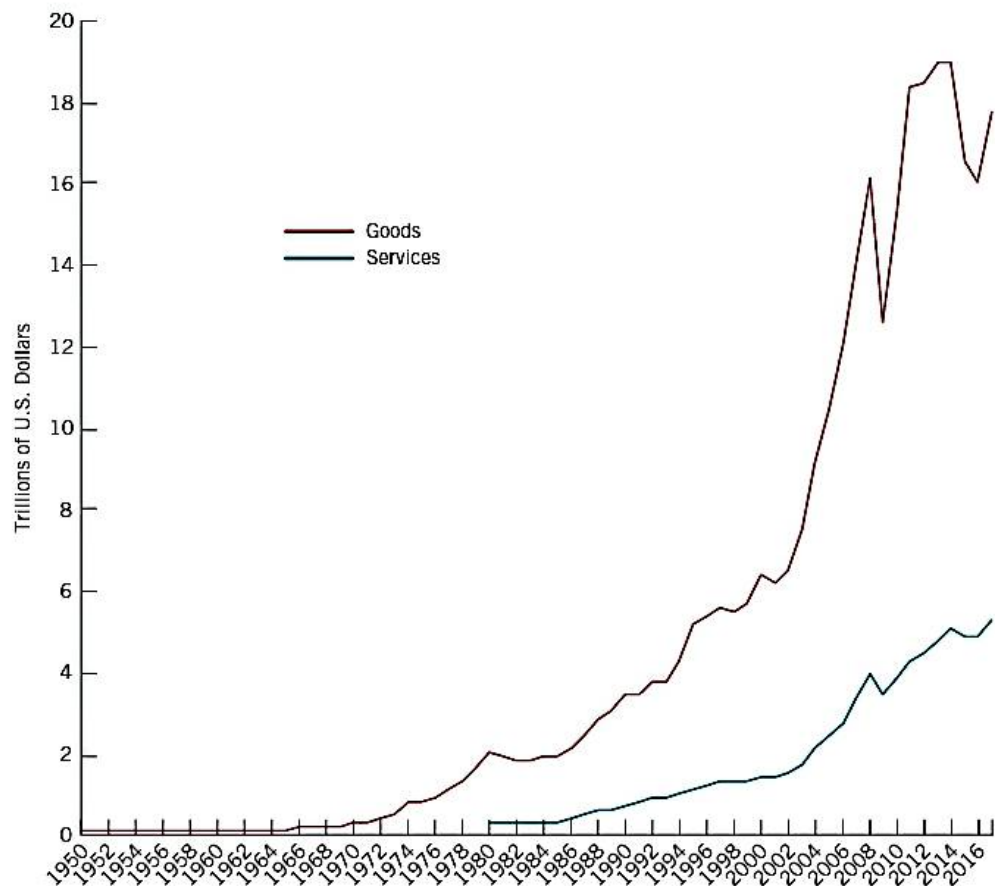
6.1. Understand the motivation for international trade.

Trade is the voluntary exchange of goods, services, assets, or money between one person or organization and another. Because it is voluntary, both parties to the transaction must believe they will gain from the exchange or else they would not complete it. **International trade** is trade between residents of two countries. The residents may be individuals, firms, not-for-profit organizations, or other forms of associations.

Why does international trade occur? The answer follows directly from our definition of trade: Both parties to the transaction, who happen to reside in two different countries, believe they benefit from the voluntary exchange. Behind this simple truth lies much economic theory, business practice, government policy, and international conflict—topics we cover in this and the next four chapters.

As Figure 6.1 indicates, world trade has grown dramatically in the more than half-century since the end of World War II. Total international merchandise trade in 2017 was \$17.7 trillion, or

FIGURE 6.1
The Growth of World Exports since 1950



Source: Based on World Trade Organization data bank, found at www.wto.org, June 2018.

approximately 22 percent of the world's \$80.7 trillion gross domestic product (GDP); trade in services that year amounted to \$5.3 trillion. The EU, the United States, Canada, and Japan accounted for 48.3 percent of the world's merchandise exports (see Figure 6.2); China, 12.8 percent; and other countries, 38.9 percent. Such international trade has important direct and indirect effects on national economies. On the one hand, exports spark additional economic activity in the domestic economy. Consider leading civilian aircraft exporters like Boeing, Textron, and Gulfstream. Their \$56.3 billion in exports in 2017 generated orders for their U.S. suppliers, wages for their U.S. workers, and dividend payments for their U.S. shareholders, all of which then create income for local automobile dealers, grocery stores, and others, which in turn add to their own payrolls.² On the other hand, imports can pressure domestic suppliers to cut their prices and improve their competitiveness. Failure to respond to foreign competition may lead to closed factories and unemployed workers.

Because of the obvious significance of international trade to businesses, consumers, and workers, scholars have attempted to develop theories to explain and predict the forces that motivate such trade. Governments use these theories when they design policies they hope will benefit their countries' industries and citizens. Managers use them to identify promising markets and profitable internationalization strategies.

In Practice

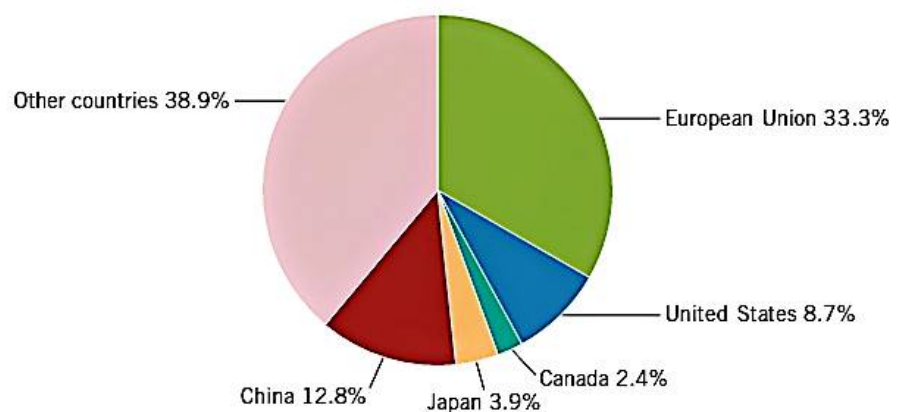
- Trade benefits both parties to the transactions. If not, why would each agree to the deal?
- International trade in goods and services has grown rapidly in the past three decades. *For further consideration:* How does international trade affect your standard of living?

6.2. Summarize and discuss the differences among the classical country-based theories of international trade.

Classical Country-Based Trade Theories

The first theories of international trade developed with the rise of the great European nation-states during the sixteenth century. Not surprisingly, these early theories focused on the individual country in examining patterns of exports and imports. As we discuss in more detail later in this chapter, these country-based theories are particularly useful for describing trade in commodities—standardized, undifferentiated goods such as oil, sugar, or lumber that are typically bought on the basis of price rather than brand name. However, as multinational enterprises (MNEs) rose to power in the middle of the twentieth century, scholars shifted their attention to the firm's role in promoting international trade. The firm-based theories developed after World War II are useful in describing patterns of trade in differentiated goods—those such as automobiles, consumer electronics, and personal care products, for which brand name is an important component of the customer's purchase decision. In this section, we examine the classical country-based theories of international trade; in the next section, we explore the more modern firm-based theories.

FIGURE 6.2
Source of the World's
Merchandise Exports, 2017



Source: Based on data from World Trade Organization (www.wto.org), June 2018.

Mercantilism

Mercantilism is a sixteenth-century economic philosophy that maintains that a country's wealth is measured by its holdings of gold and silver. According to mercantilists, a country's goal should be to enlarge these holdings by promoting exports and discouraging imports. The logic was transparent to sixteenth-century policymakers: If foreigners buy more goods from you than you buy from them, then the foreigners have to pay you the difference in gold and silver, enabling you to amass more treasure. Mercantilist terminology is still used today, for example, when television commentators and newspaper headlines report that a country suffered an "unfavorable" balance of trade—that is, its exports were less than its imports.

At the time mercantilism seemed to be sound economic policy. Large gold and silver holdings meant the reigning monarchs could afford to hire armies to fight other countries and thereby expand their realms. Politically, mercantilism was popular with many manufacturers and their workers. Export-oriented manufacturers favored mercantilist trade policies, such as those establishing subsidies or tax rebates, which stimulated sales to foreigners. Domestic manufacturers threatened by foreign imports endorsed mercantilist trade policies, such as those imposing tariffs or quotas, which protected the manufacturers from foreign competition. These businesses, their workers, their suppliers, and the local politicians representing the communities in which the manufacturers had production facilities all praised the wisdom of the monarchs' mercantilist policies.

Most members of society, however, are hurt by such policies. Governmental subsidies of the exports of certain industries are paid by taxpayers in the form of higher taxes. Governmental import restrictions are paid for by consumers in the form of higher prices because domestic firms face less competition from foreign producers. During the age of imperialism, governments often shifted the burden of mercantilist policies onto their colonies. For example, under the Navigation Act of 1660 all European goods imported by the American colonies had to be shipped from Great Britain. The British prohibited colonial firms from exporting certain goods that might compete with those from British factories, such as hats, finished iron goods, and woollens. To ensure adequate supplies of low-cost inputs for British merchants, the British required some colonial industries to sell their output only to British firms. This output included rice, tobacco, and naval stores (forest products used in shipbuilding).³ This particular mercantilist strategy ultimately backfired—it contributed to the grievances that led to the overthrow of the British Crown in the American colonies.

Because mercantilism does benefit certain members of society, mercantilist policies are still politically attractive to some firms and their workers. Modern supporters of such policies, called **neomercantilists** or **protectionists**, include such diverse U.S. groups as the American Federation of Labor–Congress of Industrial Organizations, textile manufacturers, steel companies, sugar growers, and peanut farmers.

Protectionist attitudes are not limited to the United States. North Americans and Europeans have long complained that Japan limits the access of foreign goods to its market. For instance, it took 40 years of negotiations before Japan grudgingly agreed in the 1990s to allow the importation of foreign rice, and even then it limited rice imports to less than 10 percent of its market. Asian and North American firms criticize the Europeans for imposing barriers against imported goods such as beef, bananas, and other agricultural products. Such finger-pointing is amply justified: Nearly every country has adopted some neomercantilist policies to protect key industries in its economy.

Absolute Advantage

Neomercantilism has superficial appeal, particularly to patriots who want to strengthen their country's economy. Why should a country not try to maximize its holdings of gold and silver? According to Adam Smith, the Scottish economist who is viewed as the father of free-market economics, mercantilism's basic problem is that it confuses the acquisition of treasure with the acquisition of wealth. In *An Inquiry into the Nature and Causes of the Wealth of Nations* (1776), Smith attacked the intellectual basis of mercantilism and demonstrated that it actually weakens a country because it robs individuals of the ability to trade freely and to benefit from voluntary exchanges. Moreover, in the process of avoiding imports at all costs, a country must squander its resources producing goods it is not suited to produce. The inefficiencies caused by mercantilism reduce the wealth of the country as a whole, even though certain special interest groups may benefit.

Smith advocated free trade among countries as a means of enlarging a country's wealth. Free trade enables a country to expand the amount of goods and services available to it by specializing

in the production of some goods and services and trading for others. But which goods and services should a country export and which should it import? To answer this question, Smith developed the theory of absolute advantage, which suggests that a country should export those goods and services for which it is more productive than other countries are and import those goods and services for which other countries are more productive than it is.

Absolute advantage can be demonstrated through a numerical example. Assume, for the sake of simplicity, that there are only two countries in the world, France and Japan; only two goods, wine and clock radios; and only one factor of production, labor. Table 6.1 shows the output of the two goods per hour of labor for the two countries. In France, one hour of labor can produce either two bottles of wine or three clock radios. In Japan, one hour of labor can produce either one bottle of wine or five clock radios. France has an absolute advantage in the production of wine: One hour of labor produces two bottles in France but only one in Japan. Japan has an absolute advantage in the production of clock radios: One hour of labor produces five clock radios in Japan but only three in France.

If France and Japan are able to trade with each other, both will be better off. Suppose France agrees to exchange two bottles of wine for four clock radios. Only 1 hour of French labor is needed to produce the two bottles of wine bound for Japan. In return, France will get four clock radios from Japan. These four clock radios would have required 1.33 hours of French labor had France produced them rather than buy them from Japan. By trading with Japan rather than producing the clock radios itself, France saves 0.33 hour of labor. France can use this freed-up labor to produce more wine, which in turn can be consumed by French citizens or traded to Japan for more clock radios. By allocating its scarce labor to produce goods for which it is more productive than Japan and then trading them to Japan, France can consume more goods than it could have done in the absence of trade.

Japan is similarly better off. Japan uses 0.8 hours of labor to produce the four clock radios to exchange for the two bottles of French wine. Producing the two bottles of wine itself would have required 2 hours of labor. By producing clock radios and then trading them to France, Japan saves 1.2 hours of labor, which can be used to produce more clock radios that the Japanese can consume themselves or trade to France for more wine.

Comparative Advantage

The theory of absolute advantage makes intuitive sense. Unfortunately, the theory is flawed. What happens to trade if one country has an absolute advantage in both products? The theory of absolute advantage incorrectly suggests that no trade would occur. David Ricardo, an early-nineteenth-century British economist, solved this problem by developing the theory of comparative advantage, which states that a country should produce and export those goods and services for which it is *relatively* more productive than other countries are and import those goods and services for which other countries are *relatively* more productive than it is.⁴

The difference between the two theories is subtle: Absolute advantage looks at absolute productivity differences; comparative advantage looks at relative productivity differences. The distinction occurs because comparative advantage incorporates the concept of opportunity cost in determining which good a country should produce. The opportunity cost of a good is the value of what is given up to obtain the good. Most of us apply the principles of comparative advantage and opportunity cost without realizing it. Consider a brain surgeon who is better at both brain surgery and lawn mowing than her neighbor's teenaged son is. If the surgeon is comparatively better at surgery than at lawn mowing, she will spend most of her time at the operating table and pay the teenager to mow her lawn. The brain surgeon behaves this way because the opportunity cost of mowing the lawn is too high: Time spent mowing is time unavailable for surgery.

TABLE 6.1 The Theory of Absolute Advantage: An Example

	Output Per Hour of Labor	
	France	Japan
Wine	2	1
Clock radios	3	5

Let us return to the example in Table 6.1 to contrast absolute and comparative advantage. Recall that France has an absolute advantage in wine and Japan has an absolute advantage in clock radios. The theory of absolute advantage says that France should export wine to Japan and Japan should export clock radios to France. As Table 6.1 shows, France also has a comparative advantage in wine: With 1 hour of labor it produces 2 times as much wine as Japan does but only 0.6 times as many clock radios. Thus, France is relatively more productive in wine. Japan has a comparative advantage in clock radios: With 1 hour of labor it produces 1.67 times as many clock radios as France does but only 0.5 times as much wine. So, Japan is relatively more productive in clock radios. The theory of comparative advantage says that France should export wine to Japan and Japan should export clock radios to France. For the example in Table 6.1, the theory of absolute advantage and the theory of comparative advantage both yield the same outcome.

Now let us change the facts. Suppose productivity stays the same in Japan but doubles in France as the result of new job training programs. Table 6.2 shows this new situation. France now can produce four bottles of wine or six clock radios per hour of labor. France now has an absolute advantage in both wine and clock radios: For each hour of labor, France can produce three more bottles of wine (four minus one) or one more clock radio (six minus five) than Japan can. According to the theory of absolute advantage, no trade should occur because France is more productive than Japan in producing both goods.

The theory of comparative advantage, on the other hand, indicates that trade should still occur. France is 4 times better than Japan is in wine production but only 1.2 times better in clock radio production. (Alternatively, Japan is only 0.25 times as good as France in wine production but 0.83 times as good in clock radio production.) France is comparatively better than Japan in wine production, whereas Japan is comparatively better than France in clock radio production.

By the theory of comparative advantage, France should export wine to Japan and Japan should export clock radios to France. If they do so, both will be better off. In the absence of trade, one bottle of wine will sell for 1.5 clock radios in France and for 5 clock radios in Japan. If Japan offers to trade 2 clock radios for one bottle of wine, France will be better off—even though France has an absolute advantage in clock radio production. Without trade, sacrificing one bottle of wine domestically would yield France only 1.5 clock radios in increased production. With trade, France could get 2 clock radios by giving up one bottle of wine to Japan. France gets more clock radios per bottle of wine given up by trading with Japan than by producing the clock radios domestically.

Japan also gains. Without trade, Japan has to give up five clock radios to get one more bottle of wine. With trade, Japan has to give up only two clock radios to obtain one more bottle. Japan gets more wine per clock radio given up by trading with France than by producing the wine domestically. Even though France has an absolute advantage in both wine and clock radio production, both countries gain from this trade. It is comparative advantage that motivates trade, not absolute advantage. For another insight into comparative advantage and the problems inherent in neomercantilism, see “Bringing the World into Focus.”

Comparative Advantage with Money

The lesson of the theory of comparative advantage is simple but powerful: You are better off specializing in what you do relatively best. Produce (and export) those goods and services you are relatively best able to produce, and buy other goods and services from people who are relatively better at producing them than you are.

Of course, Table 6.1 and 6.2 are both simplistic and artificial. The world economy produces more than two goods and services and is made up of more than two countries. Barriers to trade may exist, someone must pay to transport goods between markets, and inputs other than labor

TABLE 6.2 The Theory of Comparative Advantage: An Example

	Output Per Hour of Labor	
	France	Japan
Wine	4	1
Clock radios	6	5

as labor, land, and natural resources, must be consumed and thus are unavailable for other uses. When we buy goods produced by foreigners, foreign resources are used, leaving the domestic factors of production available to be used for other purposes. That, of course, is the benefit of free trade in the eyes of proponents like Adam Smith and David Ricardo: it allows a country to allocate its domestic resources to produce goods for which the country has a comparative advantage.

Thus, a more complete and appropriate analysis would be: "When we buy goods from domestic producers, we (or our countrymen) get the goods and the money, but to do so we have to consume some of our country's resources that could have been used to make other goods. When we buy goods from foreign producers, we get the goods and we get to keep the resources that would otherwise be needed to produce those goods domestically; the foreigners get our money, but they have to consume their resources to make the goods they sold to us." If we use those freed resources to specialize in the production of goods for which we have a comparative advantage—and so do our foreign trading partners—both we and our foreign trading partners will have more goods to consume after we engage in international trade.

The gains from specialization and trade are reflected in how most of us spend our daily lives. As producers, we spend our workweek specializing in providing a single service, whether it is unclogging a drain as a plumber or writing code as a computer programmer. As consumers, we buy almost everything we consume from other specialists. To obtain food, we trade money for groceries from the supermarket; to obtain clothing, we trade money for jeans or shirts from the clothing retailer. It is certainly true that when we purchase our food from the grocer, we get the food and the grocer gets our money, but both of us benefit from this transaction. If we had to grow our own food, we would have to devote more of our time and energies to that task and less time to our own productive specialty, be it computer programming or plumbing. Most of us quickly determine that our standard of living will be higher if we focus our energies on our productive specialty and then trade for goods and services produced by others. This concept applies to countries as well as individuals. By specializing in products for which they have a comparative advantage and trading freely, countries' productivity increases and, therefore, their citizens have more to consume and a higher standard of living than they would if trade were restricted by tariffs, quotas, and other barriers.

TABLE 6.3 The Theory of Comparative Advantage with Money: An Example

	Cost of Goods in France		Cost of Goods in Japan	
	French Made	Japanese Made	French Made	Japanese Made
Wine	€ 3	€ 8	¥375	¥1,000
Clock radios	€ 2	€ 1.6	¥250	¥200

Note: For example, one hour's worth of French labor can produce four bottles of wine at a total cost of €12 or an average cost of €3 per bottle. At an exchange rate of 125 yen per euro, a bottle of French-made wine will cost ¥375 ($375 = 3 \times 125$).

Given these assumptions, in the absence of trade, a bottle of wine in France costs €3, the equivalent of ¥375, and clock radios cost €2, the equivalent of ¥250. In Japan a bottle of wine costs ¥1,000 (€8), and clock radios cost ¥200 (€1.60).

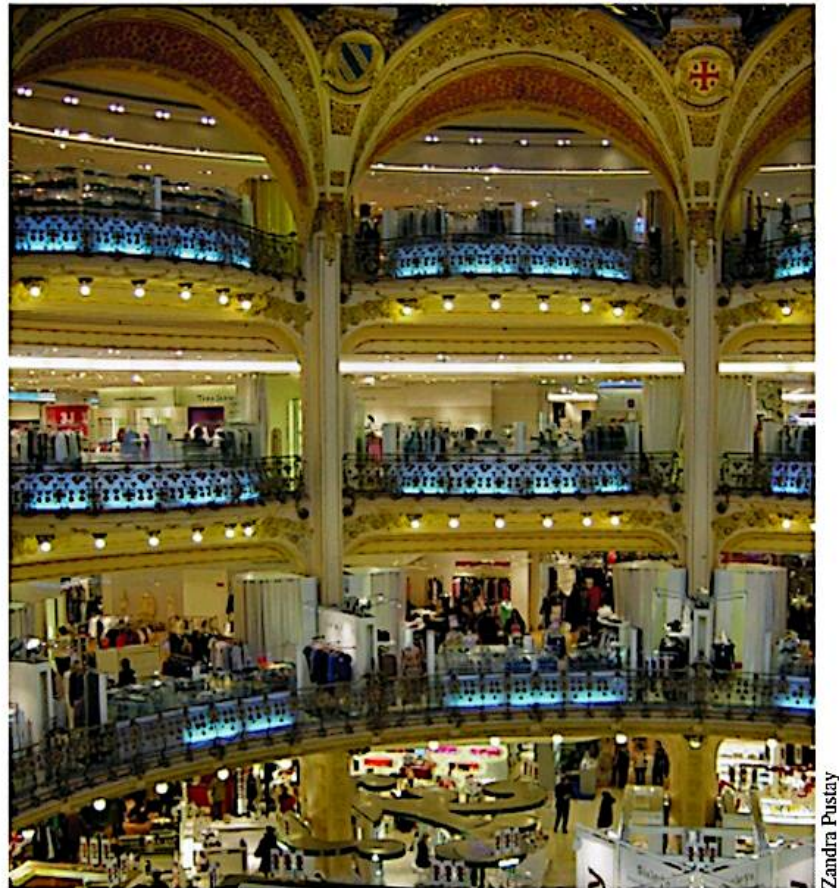
In this case, trade will occur because of the self-interest of individual entrepreneurs (or the opportunity to make a profit) in France and Japan. Suppose buyers for Galeries Lafayette, a major Paris department store, observe that clock radios cost €2 in France and the equivalent of only €1.60 in Japan. To keep their cost of goods low, these buyers will acquire clock radios in Japan, where they are cheap, and sell them in France, where they are expensive. Accordingly, clock radios will be exported by Japan and imported by France, just as the law of comparative advantage predicts. Similarly, wine distributors in Japan observe that a bottle of wine costs ¥1,000 in Japan but the equivalent of only ¥375 in France. To keep their cost of goods as low as possible, buyers for Japanese wine distributors will buy wine in France, where it is cheap, and sell it in Japan, where it is expensive. Wine will be exported by France and imported by Japan, as predicted by the law of comparative advantage.

Note that none of these businesspeople needed to know anything about the theory of comparative advantage. They merely looked at the price differences in the two markets and made their business decisions based on the desire to obtain supplies at the lowest possible cost. Yet they benefit from comparative advantage because prices set in a free market reflect a country's comparative advantage.

Relative Factor Endowments

The theory of comparative advantage begs a broader question: What determines the products for which a country will have a comparative advantage? To answer this question, two Swedish economists, Eli Heckscher and Bertil Ohlin, developed the theory of relative factor endowments,

The elegant Galeries Lafayette department store on Boulevard Haussmann, a Parisian fixture for over a century, is a magnet for international shoppers desiring the latest in fashion and design from around the world.



Zandra Pustay

now often referred to as the **Heckscher-Ohlin theory**. These economists made two basic observations:

1. *Factor endowments (or types of resources) vary among countries.* For example, Argentina has much fertile land, Saudi Arabia has large crude oil reserves, and Bangladesh has a large pool of unskilled labor.
2. *Goods differ according to the types of factors that are used to produce them.* For example, wheat requires fertile land, oil production requires crude oil reserves, and apparel manufacturing requires unskilled labor.

From these observations, Heckscher and Ohlin developed their theory: A country will have a comparative advantage in producing products that intensively use resources (factors of production) it has in abundance. Thus, Argentina has a comparative advantage in wheat growing because of its abundance of fertile land; Saudi Arabia has a comparative advantage in oil production because of its abundance of crude oil reserves; and Bangladesh has a comparative advantage in apparel manufacture because of its abundance of unskilled labor.

The Heckscher-Ohlin theory suggests a country should export those goods that intensively use those factors of production that are relatively abundant in the country. The theory was tested empirically after World War II by economist Wassily Leontief using input-output analysis, a mathematical technique for measuring the interrelationships among the sectors of an economy. Leontief believed the United States was a capital-abundant and labor-scarce economy. Therefore, according to the Heckscher-Ohlin theory, he reasoned that the United States should export capital-intensive goods, such as bulk chemicals and steel, and import labor-intensive goods, such as clothing and footwear.

Leontief used his input-output model of the economy to estimate the quantities of labor and capital needed to produce “bundles” of U.S. exports and imports worth \$1 million in 1947 (see Figure 6.3). (Each bundle was a weighted average of all U.S. exports or imports in 1947.)

6.3. Use the modern firm-based theories of international trade to describe global strategies adopted by businesses.

Modern Firm-Based Trade Theories

Since World War II, international business research has focused on the role of the firm rather than the country in promoting international trade. Firm-based theories have developed for several reasons: (1) the growing importance of MNCs in the postwar international economy; (2) the inability of the country-based theories to explain and predict the existence and growth of intra-industry trade (defined in the next section); and (3) the failure of Leontief and other researchers to empirically validate the country-based Heckscher-Ohlin theory. Unlike country-based theories, firm-based theories incorporate factors such as quality, technology, brand names, and customer loyalty into explanations of trade flows. Because firms, not countries, are the agents for international trade, the newer theories explore the firm's role in promoting exports and imports.

Product Life Cycle Theory

Product life cycle theory, which originated in the marketing field to describe the evolution of marketing strategies as a product matures, was modified by Raymond Vernon of the Harvard Business School to create a firm-based theory of international trade (and, as we will see, of international investment). International product life cycle theory traces the roles of innovation, market expansion, comparative advantage, and strategic responses of global rivals in international production, trade, and investment decisions. According to Vernon's theory, and as illustrated in Figure 6.4, the international product life cycle consists of three stages: new product, maturing product, and standardized product.

In stage 1, the *new product stage*, a firm develops and introduces an innovative product, such as a photocopier or a personal computer, in response to a perceived need in the domestic market. Because the product is new, the innovating firm is uncertain whether a profitable market for the product exists. The firm's marketing executives must closely monitor customer reactions to ensure that the new product satisfies consumer needs. Quick market feedback is important, so the product is likely to be initially produced in the country where its research and development occurred, typically a developed country such as Japan, Germany, or the United States. Further, because the market size also is uncertain, the firm usually will minimize its investment in manufacturing capacity for the product. Most output initially is sold in the domestic market, and export sales are limited.

For example, during the early days of the personal computer industry the small producers that populated the industry had their hands full trying to meet the burgeoning demand for their product. Apple Computer typified this problem. Founded on April Fool's Day in 1976, its initial assembly plant was located in the garage of cofounder Steve Jobs. The first large order for its homemade computers—50 units from a local computer hobbyist store—almost bankrupted the firm because it lacked the financing to buy the necessary parts.⁶ Apple survived because of the nurturing environment in which it was born, California's Silicon Valley. Home to major electronics firms such as Hewlett-Packard, Intel, and National Semiconductor, Silicon Valley was full of electrical engineers who could design and build Apple's products and venture capitalists who were seeking the "next Xerox." It was the perfect locale for Apple's sales to grow from zero in 1976 to \$7.8 million in 1978 and \$229 billion in 2017.⁷

In stage 2, the *maturing product stage*, demand for the product expands dramatically as consumers recognize its value. The innovating firm builds new factories to expand its capacity and satisfy domestic and foreign demand for the product. Domestic and foreign competitors begin to emerge, lured by the prospect of lucrative earnings. In the case of Apple, the firm introduced a hand-assembled version of its second model, the Apple II, at a San Francisco computer fair in the spring of 1977. Within three years Apple had sold 130,000 units and expanded its production facilities beyond Jobs' garage. To serve domestic and foreign customers, Apple IIs were manufactured in California and Texas and distributed from warehouses in the United States and the Netherlands.

In stage 3, the *standardized product stage*, the market for the product stabilizes. The product becomes more of a commodity, and firms are pressured to lower their manufacturing costs as much as possible by shifting production to facilities in countries with low labor costs. As a result, the product begins to be imported into the innovating firm's home market (by either the firm or its competitors). In some cases, imports may result in the complete elimination of domestic production.

The personal computer industry has entered the standardized product stage. Chinese computer companies like Lenovo and Taiwanese original design manufacturers such as Compal, Quanta,

Wistron, Inventec, and Hon Hai—none of them household names in the United States—annually export to the United States and to other countries millions of personal computers, many of which are produced under contract for leading PC vendors like Apple, Dell, and Hewlett-Packard. Such an arrangement is typical in this stage of the product life cycle: the production of the good shifts to lower-cost manufacturing sites, but the branding and marketing functions remain in the innovating country.⁸

According to the international product life cycle theory, domestic production begins in stage 1, peaks in stage 2, and slumps in stage 3. Exports by the innovating firm's country also begin in stage 1 and peak in stage 2. By stage 3, however, the innovating firm's country becomes a net importer of the product. Foreign competition begins to emerge toward the end of stage 1, as firms in other industrialized countries recognize the product's market potential. In stage 2, foreign competitors expand their productive capacity, thus servicing an increasing portion of their home markets and perhaps becoming net exporters. As competition intensifies in stage 2, however, the innovating firm and its domestic and foreign rivals seek to lower their production costs by shifting production to low-cost sites in less-developed countries. Eventually, in stage 3, the less-developed countries may become net exporters of the product.

Country Similarity Theory

Country-based theories, such as the theory of comparative advantage, do a good job of explaining interindustry trade among countries. **Interindustry trade** is the exchange of goods produced by one industry in country A for goods produced by a different industry in country B, such as the exchange of French wines for Japanese clock radios. Yet much international trade consists of **intraindustry trade**, that is, trade between two countries of goods produced by the same industry. For example, Japan exports Toyotas to Germany, and Germany exports BMWs to Japan. Such intraindustry trade is not predicted by country-based theories.

Swedish economist Steffan Linder sought to explain the phenomenon of intraindustry trade. Linder hypothesized that international trade in manufactured goods results from similarities of preferences among consumers in countries that are at the same stage of economic development. In his view, firms initially manufacture goods to serve the firms' domestic market. As they explore exporting opportunities, they discover that the most promising foreign markets are in countries where consumer preferences resemble those of their own domestic market. The Japanese market, for example, provides BMW with well-off, prestige- and performance-seeking automobile buyers similar to the ones who purchase its cars in Germany. The German market provides Toyota with quality-conscious and value-oriented customers similar to those found in its home market. As each company targets the other's home market, intraindustry trade arises. Linder's **country similarity theory** suggests that most trade in manufactured goods should be between countries with similar per capita incomes and that intraindustry trade in manufactured goods should be common. This theory is particularly useful in explaining trade in **differentiated goods** such as automobiles, expensive electronics equipment, and personal care products, for which brand names and product reputations play an important role in consumer decision making. (**Undifferentiated goods**, such as coal, petroleum products, and sugar, are those for which brand names and product reputations play a minor role at best in consumer purchase decisions.)

New Trade Theory

The so-called **new trade theory** developed by Elhanan Helpman, Paul Krugman,⁹ and Kelvin Lancaster,¹⁰ in the 1970s and 1980s extends Linder's analysis by incorporating the impact of economies of scale on trade in differentiated goods. **Economies of scale** occur if a firm's average costs of producing a good decrease as its output of that good increases. Suppose automobile manufacturing and marketing benefits from economies of scale. If both BMW and Toyota are able to expand their sales beyond their home market, each will benefit from lower average costs and each will improve its competitiveness vis-à-vis smaller firms. In industries in which economies of scale are important, we would expect firms to be particularly aggressive in expanding beyond their domestic markets.

Like Linder's approach, the new trade theory predicts that intraindustry trade will be commonplace. It also suggests MNCs within the same industry, such as Caterpillar and Komatsu, Unilever and Procter & Gamble, and Airbus and Boeing, will continually play cat-and-mouse games with one another on a global basis as they attempt to expand their sales to capture scale

economies. Often they seek to harness some sustainable competitive advantage they enjoy as a means of leveraging their own strengths and neutralizing those of their rivals.

Firms competing in the global marketplace have numerous ways of obtaining a sustainable competitive advantage. The more popular ones are owning intellectual property rights, investing in research and development (R&D), achieving economies of scope, and exploiting the experience curve. We discuss each of these options next.

OWNING INTELLECTUAL PROPERTY RIGHTS A firm that owns an intellectual property right—a trademark, brand name, patent, or copyright—often gains advantages over its competitors. For instance, owning prestigious brand names enables Switzerland's Rolex SA and France's LVMH Moët Hennessy Louis Vuitton to charge premium prices for their upscale products. And Coca-Cola and PepsiCo compete for customers worldwide on the basis of their trademarks and brand names.

INVESTING IN R&D R&D is a major component of the total cost of high-technology products. For example, Airbus has spent more than \$25 billion developing its superjumbo jet, the A380.¹¹ Firms in the computer, pharmaceutical, and semiconductor industries also spend large amounts on R&D to maintain their competitiveness. Because of such large “entry” costs, other firms often hesitate to compete against established firms. Thus, the firm that acts first often gains a first-mover advantage.

However, knowledge does not have a nationality. Firms that invest up front and secure the first-mover advantage have the opportunity to dominate the world market for goods that are intensive in R&D. Thus, national competitiveness and trade flows may be determined by which firms make the necessary R&D expenditures. Why is the EU a large exporter of commercial aircraft? Because Airbus is one of the few firms willing to spend the large sums of money required to develop new aircraft and because it is headquartered in Europe.

Firms with large domestic markets may have an advantage over their foreign rivals in high-technology markets because these firms often are able to obtain quicker and richer feedback from customers. With this feedback the firms can fine-tune their R&D efforts, enabling the firms to better meet the needs of their domestic customers. This knowledge can then be used to serve foreign customers. For example, U.S. agricultural chemical producers such as Monsanto and Eli Lilly have an advantage over Japanese rivals in developing soybean pesticides because the U.S. market for such pesticides is large whereas the Japanese market is small. Knowledge gained in the U.S. pesticide market can be readily transferred to meet the needs of Japanese farmers.

ACHIEVING ECONOMIES OF SCOPE Economies of scope offer firms another opportunity to obtain a sustainable competitive advantage in international markets. Economies of scope occur when a firm's average costs decrease as the number of different products it sells increases. Firms that are able to achieve economies of scope enjoy low average costs, which give the firms a competitive advantage over their global rivals. Consider the e-retailer Amazon, which has benefited from both economies of scale and scope. It has spent enormous sums developing and maintaining its website and building its customer base. Because many of these costs are fixed, the company's average costs per sale decline as the company expands its sales. In its quest to capture the volume-driven economies of scale, Amazon has been expanding its operations into the international marketplace. Moreover, the marginal cost of adding an additional product line to its website is relatively small. Accordingly, Amazon has expanded from books to compact discs to DVDs to a myriad of other goods to capture such economies of scope.

EXPLOITING THE EXPERIENCE CURVE Another source of firm-specific advantages in international trade is exploitation of the experience curve. For certain types of products, production costs decline as the firm gains more experience in manufacturing the product. Experience curves may be so significant that they govern global competition within an industry. For instance, in semiconductor chip production, unit cost reductions of 25 to 30 percent with each doubling of a firm's cumulative chip production are not uncommon.¹² Any firm attempting to be a low-cost producer of so-called commodity chips—such as DRAM memory chips—can achieve that goal only if it moves further along the experience curve than its rivals do. Both U.S. and Asian chip manufacturers have often priced their new products below current production costs to capture the sales necessary to generate the production experience that will in turn enable the manufacturers to lower future production costs. Because of their technological leadership in manufacturing and their aggressive, price-cutting strategies, Asian semiconductor manufacturers such as Samsung, Taiwan Semiconductor Manufacturing, and SK Hynix dominate the production of low-cost,

standardized semiconductor chips.¹³ Similarly, innovative U.S. semiconductor firms such as Intel and Qualcomm use the experience curve to maintain leadership in the production of high-priced, proprietary chips that form the brains of newer microcomputers and smartphones.

Porter's Theory of National Competitive Advantage

Harvard Business School professor Michael Porter's **theory of national competitive advantage** is the newest addition to international trade theory. Porter believes that success in international trade comes from the interaction of four country- and firm-specific elements: factor conditions; demand conditions; related and supporting industries; and firm strategy, structure, and rivalry.

FACTOR CONDITIONS A country's endowment of factors of production affects its ability to compete internationally. Although factor endowments were the centerpiece of the Heckscher-Ohlin theory, Porter goes beyond the basic factors—land, labor, and capital—considered by the classical trade theorists to include more advanced factors such as the educational level of the workforce and the quality of the country's infrastructure. His work stresses the role of factor creation through training, research, and innovation.

DEMAND CONDITIONS The existence of a large, sophisticated domestic consumer base often stimulates the development and distribution of innovative products as firms struggle for dominance in their domestic markets. In meeting their domestic customers' needs, however, firms continually develop and fine-tune products that also can be marketed internationally. Thus, pioneering firms can stay ahead of their international competitors as well. For example, Japanese consumer electronics producers benefit internationally because of the willingness of Japan's large, well-off middle class to buy the latest electronic creations of Sony, Toshiba, and Matsushita. After being fine-tuned in the domestic market, new models of Japanese digital cameras, big-screen TVs, and Blu-ray players are sold to eager European and North American consumers. A similar phenomenon is occurring in the telecommunications market, where the rapid adoption of the Internet, tablets, and smartphones by North American consumers and companies has created a fertile climate for companies such as Twitter, Facebook, eBay, and Amazon to develop and tailor new products to meet the needs of this market domestically and internationally.

RELATED AND SUPPORTING INDUSTRIES The emergence of an industry often stimulates the development of local suppliers eager to meet that industry's production, marketing, and distribution needs. An industry located close to its suppliers will enjoy better communication and the exchange

The Netherlands' rich farmland and favorable climate makes it the world's largest producer of tulip and daffodil bulbs—an estimated 9 billion per year. Dutch producers in turn benefit from sophisticated supporting industries and trade associations, such as the Dutch Flower Council, the Association of Dutch Flower Growers Research Groups, and the FloraHolland flower auctions, which facilitate the competitiveness of Dutch flower growers.



Scott Pustay

of cost-saving ideas and inventions with those suppliers. Competition among these input suppliers leads to lower prices, higher-quality products, and technological innovations in the input market, in turn reinforcing the industry's competitive advantage in world markets. For example, Hollywood's dominance of the world film industry is based in part on the local availability of specialist input suppliers, such as casting directors, stunt coordinators, costume and set designers, demolition experts, animators, special-effects firms, and animal wranglers.

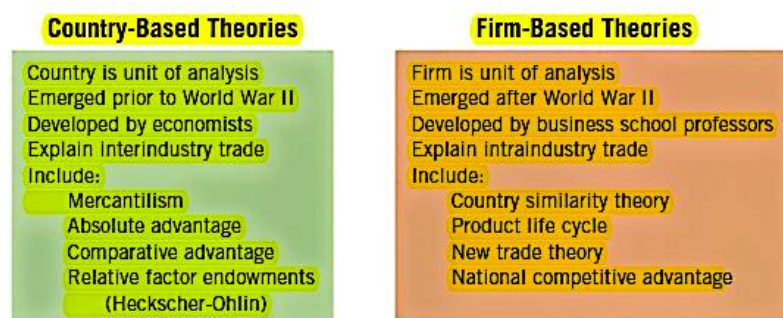
FIRM STRATEGY, STRUCTURE, AND RIVALRY The domestic environment in which firms compete shapes their ability to compete in international markets. To survive, firms facing vigorous competition domestically must continuously strive to reduce costs, boost product quality, raise productivity, and develop innovative products. Firms that have been tested in this way often develop the skills needed to succeed internationally. Further, many of the investments they have made to succeed in the domestic market (for example, in R&D, quality control, brand image, and employee training) are transferable to international markets at low cost. Such firms have an edge as they expand abroad. Thus, according to Porter's theory, the international success of Japanese automakers and consumer electronics manufacturers and of Hollywood film studios is aided by intense domestic competition in these firms' home countries.

Porter holds that national policies may also affect firms' international strategies and opportunities in more subtle ways. Consider the German automobile market. German labor costs are high, so German automakers find it difficult to compete internationally on the basis of price. As most auto enthusiasts know, however, there are no speed limits on many stretches of Germany's famed autobahns. So German automakers such as Daimler-Benz, Porsche, and BMW have chosen to compete on the basis of quality and high performance by engineering chassis, engines, brakes, and suspensions that can withstand the stresses of high-speed driving. Consequently, these firms dominate the world market for high-performance automobiles.

Porter's theory is a hybrid: It blends the traditional country-based theories that emphasize factor endowments with the firm-based theories that focus on the actions of individual firms. Countries (or their governments) play a critical role in creating an environment that can aid or harm the ability of firms to compete internationally, but firms are the actors that actually participate in international trade. Some firms succeed internationally; others do not. Porsche, Daimler-Benz, and BMW successfully grasped the opportunity presented by Germany's decision to allow unlimited speeds on its highways and captured the high-performance niche of the worldwide automobile industry. Conversely, Opel chose to focus on the broader middle segment of the German automobile market, ultimately limiting its international options.

In summary, no single theory of international trade explains all trade flows among countries. The classical country-based theories are useful in explaining interindustry trade of homogeneous, undifferentiated products, such as agricultural goods, raw materials, and processed goods like steel and aluminum. The firm-based theories are more helpful in understanding intraindustry trade of heterogeneous, differentiated goods, such as Samsung smartphones and Caterpillar bulldozers, many of which are sold on the basis of their brand names and reputations. Further, in many ways, Porter's theory synthesizes the features of the existing country-based and firm-based theories. It also explains another important phenomenon of international competition, the regional clustering of firms within an industry, which is discussed in "Venturing Abroad." Figure 6.5 summarizes the major theories of international trade.

FIGURE 6.5
Theories of
International Trade



VENTURING ABROAD



BIRDS OF A FEATHER FLOCK TOGETHER

Although important for the development of trade theory, Porter's theory of national competitive advantage also explains the existence of another phenomenon of international business activity, the regional **clustering** of firms within an industry resulting from agglomeration economies. **Agglomeration economies** occur when a firm's costs of production decline as the number of firms in that industry increase within a given area. Such growth attracts additional input suppliers to the areas, which then increases price competition and innovation among those suppliers. As their customer base increases, suppliers can specialize, developing unique abilities that benefit the cluster as a whole. Clusters also promote innovation and entrepreneurship. Competition is intense, and firms must continually improve their products and productivity to survive. Entrepreneurs can tap into sophisticated local knowledge networks. Moreover, local bankers and financiers understand the ins-and-outs of the local industry and thus are better able to recognize good ideas when entrepreneurs request loans or capital. Firms within the cluster thus enjoy significant advantages when competing with a firm from outside the cluster. Map 6.1 depicts some key industrial clusters in Western Europe.

Consider three California industries, filmed entertainment, centered in Los Angeles; premium wines, centered in Napa and Sonoma counties; and Internet software services, centered in the Silicon Valley. Each industry no doubt started and benefited from supportive factor endowments and demand conditions. But as area firms began to prosper, other firms within the same industry were attracted to the region seeking to replicate the pioneering firms' success. The expanding number of customers then induced supplier firms to relocate as well. Over time, the cluster becomes so strong that firms not in the cluster are at a significant disadvantage. Film studios requiring the best directors, cinematographers, screenwriters, casting agents, and such are likely to find them in Hollywood. Suppliers of specialized services like pyrotechnics, animal wrangling, special effects, and set

design are readily available as well. Similarly, firms seeking the latest vineyard management techniques or viticulture science are likely to find them in Napa or Sonoma counties or at the nearby University of California at Davis. Firms specializing in making or supplying wine-making and grape-harvesting equipment, barrels, corks, bottles, and label design have proliferated there as well, benefiting and strengthening the area's vineyards. Or consider the move of Facebook from Cambridge, Massachusetts, to the Silicon Valley as portrayed in the movie *The Social Network*. There are plenty of smart people in Cambridge. However, Facebook founder Mark Zuckerberg realized that Facebook needed to access the specialized resources and talent that the Silicon Valley could best provide if it were to dominate the social networking market.

Porter argues that clusters play an important role in promoting international competitiveness. Such competition is being transformed from a firm-versus-firm basis to a cluster-versus-cluster basis. In wine, for example, we often think of competition as being between a French vineyard such as Chateau Lafite Rothschild or a California grower like Chateau Montelena. From a cluster perspective, however, the nature of competition changes: California's Sonoma and Napa Valley vintners compete against vintners from France's Burgundy and Champagne provinces and from growers in Australia's Barossa Valley. In Porter's view, a wise government institutes policies that allow the cluster to flourish, perhaps by funding research at local universities or providing infrastructure improvements that benefit the cluster as a whole.

Sources: Based on Harvard Business School, "Global Wine War 2009: New World versus Old," Case number 9-910-405 (2009); Harvard Business School, "Finland and Nokia: Creating the World's Most Competitive Economy," Case number 9-702-427 (2008); Michael Porter, "Clusters and the New Economics of Competition," *Harvard Business Review*, November–December 1998, pp. 77–90; Michael Porter, "The Competitive Advantage of Nations," *Harvard Business Review*, March–April 1990, pp. 73–93.

In Practice

- The firm-based trade theories highlight the importance of multinational corporations in international trade and international investment flows.
- Regional clustering of firms within an industry can be an important source of competitive advantage to firms and to countries.

For further consideration: What percentage of your budget do you spend on undifferentiated goods? On differentiated goods?

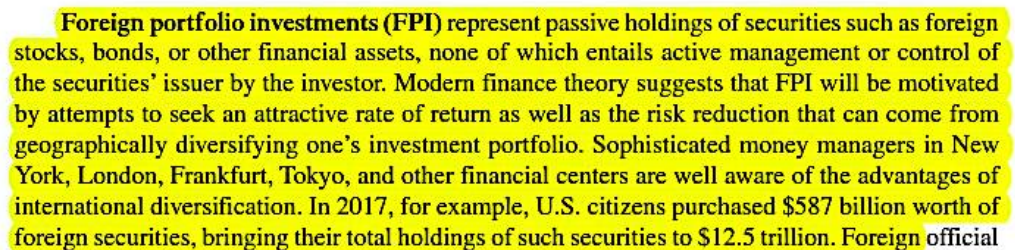
An Overview of International Investment

6.4. Describe and categorize the different forms of international investment.

Trade is the most obvious but not the only form of international business. Another major form is international investment, whereby residents of one country supply capital to a second country.

Types of International Investments

International investment, as discussed in Chapter 1, is divided into two categories: foreign portfolio investment (FPI) and foreign direct investment (FDI). The distinction between the two rests on the question of control: Does the investor seek an active management role in the firm or merely a return from a passive investment?



and private investors purchased \$799 billion worth of U.S. corporate, federal, state, and local securities, raising their total holdings of such securities to \$19.5 trillion.¹⁴

Foreign direct investment (FDI) is acquisition of foreign assets for the purpose of controlling them. U.S. government statisticians define FDI as “ownership or control of 10 percent or more of an enterprise’s voting securities or the equivalent interest in an unincorporated business.”¹⁵ FDI may take many forms, including purchase of existing assets in a foreign country, new investment in property, plant, and equipment, and participation in a joint venture with a local partner. Perhaps the most historically significant FDI in the United States was the \$24 that Dutch explorer Peter Minuit paid local Native Americans for Manhattan Island.¹⁶ The result: New York City, one of the world’s leading financial and commercial centers. “Venturing Abroad” discusses the growing importance of sovereign wealth funds in global capital markets.

VENTURING ABROAD



THE NEW PLAYER IN GLOBAL CAPITAL MARKETS: SOVEREIGN WEALTH FUNDS

Sovereign wealth funds (SWFs) are one of the newest, and perhaps most controversial, sources of capital in the world economy. Sovereign wealth funds, which are government-controlled pools of capital, are estimated to possess \$7.9 trillion in assets in 2018.

Most of the SWFs are owned by oil-rich governments. Norway’s Government Pension Fund is currently the largest, with \$1 trillion in assets, followed by the China Investment Corporation with \$941 billion. Other SWFs owned by governments of countries enjoying large oil revenues include the Abu Dhabi Investment Authority (\$683 billion), the Kuwait Investment Authority (\$592 billion), and Saudi Arabia’s SMA Foreign Holdings (\$494 billion).

The creation of the China Investment Corporation (CIC) was the result of the large foreign exchange reserves (totaling some \$3.4 trillion) the country acquired as a result of its balance of payments surpluses. The government transferred \$200 billion of these reserves in 2007 to the CIC to manage in hopes of improving the government’s rate of return on its investments. One of the first investments made by the CIC was a \$3 billion equity position in the Blackstone Group, a U.S. private equity firm.

The rise of the SWFs reflects the growing role of emerging markets in the global economy. Their ownership of the world’s productive assets has grown in conjunction with their increasing share of the world’s GDP. SWFs have invested in a variety of industries. For example, Dubai International Capital, owned by the ruler of Dubai, owns stakes in the Airbus Group (the parent of Airbus), Sony, and HSBC Holdings. It has also purchased ownership stakes in the London Stock Exchange, the Nasdaq Stock Market, and OMX AB, which controls seven stock exchanges in Scandinavia and the Baltic region. The Qatar Investment Authority holds major positions in Volkswagen, Porsche, Total, Barclays, J Sainsbury, and Harrods. Similarly, Temasek purchased a controlling interest in Shin Corporation, the largest provider of mobile telephone service in Thailand.

The SWFs were particularly important in shoring up the balance sheets of international banks, hedge funds, and private equity firms hurt by the chaos in the U.S. mortgage market during the Global Recession of 2007–2009. UBS, which lost \$37 billion in the crisis, received \$9.7 billion in new equity from the Government of Singapore Investment Corporation, and an additional \$1.8 billion from an unidentified strategic investor from the Middle East. The Singapore SWF owns about 9 percent of the Swiss bank as a result of this investment. The China Investment Corporation invested \$5 billion in Morgan Stanley, while the Abu Dhabi Investment Authority purchased

a 4.9 percent stake in Citigroup for \$7.5 billion and a 7.5 percent share of the Carlyle Group for \$1.35 billion.

Most of the investments made by SWFs to date have been passive in nature, seeking to earn financial returns rather than actively control assets. Many have purchased equity stakes in investment banks and private equity firms, allowing the SWFs to tap into the sophisticated knowledge of these concerns without having to develop the expertise themselves. They are also attractive to the firms because to date they have not concerned themselves with corporate governance issues or proven to be activist investors. Their investments typically do not entitle the SWF to a board seat, allowing them to avoid political problems and national investment control laws like the Exon-Florio Act. (This act allows the president of the United States to reject certain investments made by foreign nationals for national security reasons.)

Nonetheless, the growth of SWFs has made politicians in many countries uneasy. As a result, the International Monetary Fund is trying to develop a voluntary code of conduct for SWFs. The primary objective of the code is to ensure that investments made by SWFs are solely for commercial purposes, rather than a backdoor means of promoting the political goals of the governmental owner of the SWF.

Sources: Based on Sovereign Wealth Funds Institute website (www.swfinstitute.org), accessed August 16, 2018; “How to not spend it,” *The Economist*, September 24, 2016; “China’s forex reserves reach \$3.4tn,” *Financial Times*, April 11, 2013; “Here, there and everywhere,” *Financial Times*, March 19, 2013, p. 7; “Norway’s national nest egg,” *Financial Times*, August 20, 2012, p. 5; “Wealth funds fight for reputation,” *Wall Street Journal*, May 10, 2011, p. C3; “Beijing to give wealth fund \$200bn,” *Financial Times*, April 26, 2011, p. 4; “China’s wealth-fund chief warns on global growth,” *Wall Street Journal*, April 18, 2011, p. A7; “IMF clears way for development of sovereign wealth fund codes,” *Wall Street Journal*, March 24, 2008, p. A12; “Code set for state-run funds,” *Wall Street Journal*, March 21, 2008, p. A4; “Temasek rising,” *TheDeal.com*, February 22, 2008 (online); “Dubai investment fund plans to spend in Asia,” *International Herald Tribune*, February 19, 2008 (online); “Ascent of sovereign-wealth funds illustrates new world order,” *Wall Street Journal*, January 28, 2008, p. A2; “Asset-backed insecurity,” *The Economist*, January 19, 2008, p. 78ff.; “Gulf states seen raising foreign-asset holdings,” *Wall Street Journal*, January 17, 2008, p. A10; “\$9.4 billion write-down at Morgan Stanley,” *New York Times*, December 20, 2007 (online); “UBS gains two new investors, writes down \$10 billion,” *Wall Street Journal*, December 10, 2007 (online); “Abu Dhabi to bolster Citigroup with \$7.5 billion capital infusion,” *Wall Street Journal*, November 27, 2007, p. A3.

3. *Internalization advantage.* The firm must benefit more from controlling the foreign business activity than from hiring an independent local company to provide the service. Control is advantageous, for example, when monitoring and enforcing the contractual performance of the local company is expensive, when the local company may misappropriate proprietary technology, or when the firm's reputation and brand name could be jeopardized by poor behavior by the local company. All of these factors are important to Caterpillar.

In Practice

- International investments can take the form of portfolio investments or of direct investments. The distinction between the two forms rests on whether the investment is for purposes of control.
- Dunning's eclectic theory argues that ownership, location, and internalization advantages determine whether firms will use FDI when entering a foreign market.

For further consideration: Does foreign direct investment affect you or your community? If so, how?

6.6. Summarize how supply, demand, and political factors influence foreign direct investment.

Factors Influencing FDI

Given the complexity of the global economy and the diversity of opportunities that firms face in different countries, it is not surprising that numerous factors may influence a firm's decision to undertake FDI. These can be classified as supply factors, demand factors, and political factors (see Table 6.5).

Supply Factors

A firm's decision to undertake FDI may be influenced by supply factors, including production costs, logistics, availability of natural resources, and access to key technology.

PRODUCTION COSTS Firms often undertake FDI to lower production costs. Foreign locations may be more attractive than domestic sites because of lower land prices, tax rates, commercial real estate rents, or because of better availability and lower cost of skilled or unskilled labor. For example, Intel built a new chip fabrication facility in Chengdu in China's inland Sichuan province because labor and land costs were much lower than in Shanghai, where the company already operates three facilities.¹⁸ Similarly, Nokia built a \$275-million mobile phone assembly plant in northern Vietnam to take advantage of the area's low labor costs.¹⁹

LOGISTICS If transportation costs are significant, a firm may choose to produce in the foreign market rather than export from domestic factories. For example, Heineken has used FDI extensively as part of its internationalization strategy because its products are primarily water. Brewing its beverages close to where its foreign consumers live is cheaper for Heineken than transporting the beverages long distances from the company's Dutch breweries. International businesses also often make host-country investments to reduce distribution costs. For example, Citrovita, a Brazilian producer of orange juice concentrate, operates a storage and distribution terminal at the Port of Antwerp rather than ship to European grocery chains directly from Brazil. Citrovita can take advantage of low ocean-shipping rates to transport its goods in bulk from Brazil to the Belgian

TABLE 6.5 Factors Affecting the FDI Decision

Supply Factors	Demand Factors	Political Factors
Production costs	Customer access	Avoidance of trade barriers
Logistics	Marketing advantages	Economic development incentives
Resource availability	Exploitation of competitive advantages	
Access to technology	Customer mobility	

port. The company then uses the Antwerp facility to repackage and distribute concentrate to its customers in France, Germany, and the Benelux countries.

AVAILABILITY OF NATURAL RESOURCES Firms may use FDI to access natural resources that are critical to their operations. For instance, because of the decrease in conventional onshore domestic oil production, many U.S.-based oil companies made significant investments worldwide to obtain new oil reserves. Often international businesses negotiate with host governments to obtain access to raw materials in return for FDI. For example, the China National Petroleum Company created a \$10 billion joint venture with state-owned Petr leos de Venezuela to extract, refine, and transport 1 million barrels of oil a day from Venezuela's Orinoco basin.²⁰

ACCESS TO KEY TECHNOLOGY Another motive for FDI is to gain access to technology. Firms may find it more advantageous to acquire ownership interests in an existing firm than to assemble an in-house group of research scientists to develop or reproduce an emerging technology. For instance, many Swiss pharmaceutical manufacturers have invested in small U.S. biogenetics companies as an inexpensive means of obtaining cutting-edge biotechnology. Similarly, Korea's Doosan Infracore paid \$4.9 billion for the Bobcat division of Ingersoll-Rand to benefit from Bobcat's superior technology, outstanding distribution network, and skilled management team.²¹ Starbucks purchased a 600-acre plot in Costa Rica as an experiment station for developing new coffee varieties and new coffee-growing secrets.²²

Demand Factors

Firms also may engage in FDI to expand the market for their products. The demand factors that encourage FDI include customer access, marketing advantages, exploitation of competitive advantages, and customer mobility.

CUSTOMER ACCESS Many types of international business require firms to have a physical presence in the market. For example, fast-food restaurants and retailers must provide convenient access to their outlets for competitive reasons. KFC cannot supply its freshly prepared fried chicken to Japanese customers from its restaurants in the United States; it must locate outlets in Japan to do so. Similarly, IKEA's success in broadening its customer base beyond its home market in Sweden is the result of its opening a number of new stores worldwide.

MARKETING ADVANTAGES FDI may generate several types of marketing advantages. The physical presence of a factory may enhance the visibility of a foreign firm's products in the host market. The foreign firm also gains from "buy local" attitudes of host country consumers. For example, through ads in such magazines as *Time* and *Sports Illustrated*, Toyota has publicized the beneficial impact of its U.S. factories and input purchases on the U.S. economy. Firms may also engage in FDI to improve their customer service. Locating production facilities or design centers close to those of major customers allows suppliers to respond quickly and flexibly to last minute changes demanded by those customers.

EXPLOITATION OF COMPETITIVE ADVANTAGES FDI may be a firm's best means to exploit a competitive advantage that it already enjoys. An owner of a valuable trademark, brand name, or technology may choose to operate in foreign countries rather than export to them. Often this decision depends on the product's nature. For instance, companies such as Procter and Gamble, Nestl , and Unilever often choose to site factories in the countries in which they sell their products. By so doing, they enhance their ability to customize their products to meet local tastes while still benefitting from the power of their brand names and manufacturing prowess.

CUSTOMER MOBILITY A firm's FDI also may be motivated by the FDI of its customers or clients. If one of a firm's existing customers builds a foreign factory, the firm may decide to locate a new facility of its own nearby, thus enabling it to continue to supply its customers promptly and attentively. Equally important, establishing a new facility reduces the possibility that a competitor in the host country will step in and steal the customer. Alabama, for instance, is the home of factories assembling Honda, Hyundai, and Mercedes-Benz vehicles. In order to meet the needs of these clients, dozens of foreign-owned parts suppliers have located their own facilities in the state to serve the assembly plants, including Germany's ZF, Spain's Gestamp Automoci n, and Korea's Seohan Auto. Their need to locate facilities in the United States is magnified by the automakers'

use of just-in-time (JIT) inventory management techniques; JIT minimizes the amount of parts inventory held at an assembly plant, putting a parts-supply facility located in Europe or Asia at a severe disadvantage. The proliferation of parts suppliers in turn raises the attractiveness of siting new assembly plants in the state. In 2018, for instance, Toyota and Mazda announced they would invest \$1.6 billion in a new assembly plant, with production scheduled to commence in 2021.²³

Political Factors

Political factors may also enter into a firm's decision to undertake FDI. Firms may invest in a foreign country to avoid trade barriers by the host country or to take advantage of economic development incentives offered by the host government.

AVOIDANCE OF TRADE BARRIERS Firms often build foreign facilities to avoid trade barriers. For instance, Harley-Davidson built a factory in India to avoid that country's tariffs and taxes on motorcycle imports, which effectively doubled their price. The company is also constructing an assembly facility in Thailand for similar reasons.²⁴ Other types of government policies may also impact FDI. Microsoft, for example, located a software development center in Richmond, British Columbia, in part to avoid limitations placed by the U.S. government on the number of highly skilled immigrant workers who can obtain H-1B work visas in any given year.²⁵

ECONOMIC DEVELOPMENT INCENTIVES Most democratically elected governments—local, state, and national—are vitally concerned with promoting the economic welfare of their citizens, many of whom are, of course, voters. Many governments offer incentives to firms to induce them to locate new facilities in the governments' jurisdictions. Governmental incentives that can be an important catalyst for FDI include reduced utility rates, employee training programs, infrastructure additions (such as new roads and railroad spurs), and tax reductions or tax holidays. Often MNCs benefit from bidding wars among communities eager to attract the companies and the jobs they bring. For instance, the state of Alabama and the city of Huntsville offered an incentive package worth \$700 million to lure Toyota and Mazda to locate their new \$1.6 billion factory there. The new assembly plant, which will produce 300,000 vehicles beginning in 2021, will create 4,000 new jobs in the area. The incentive package included property tax abatements, investment credits, and construction of a \$20 million training facility.²⁶